

Institute of Actuaries of India

Subject CM2A – Financial Engineering and Loss Reserving (Paper A)

May 2023 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**i) Technical analysis**

- Technical analysis is purely based on existing historical data [0.5]
- If a market is efficient in the weak sense, technical analysis fails in generating excess risk-adjusted returns since prices already reflect this information. [0.5]
- If technical analysis enables investors to generate excess risk adjusted return then strong form efficient. It is therefore inefficient. [1]

Fundamental analysis

- Fundamental analysis uses publicly available information. [0.5]
- In a strong form efficient market, fundamental analysis does not help investors to generate excess risk adjusted return. [0.5]
- It works only in a semi-strong form efficient market, where prices and trades are already based on publicly available information. [1]

[4]**ii)****a)**

- 1) First order stochastic dominance – This theorem states that A will be preferred to B if the investor prefers more to less, $U'(x) > 0$ & $F_A(x) \leq F_B(x)$ for all x with $F_A(x) < F_B(x)$ for at least one x.
- 2) Second order stochastic dominance- This theorem states that A will be preferred to B if the investor prefers more to less, $U'(x) > 0$ & $U''(x) < 0$ and $\int_a^x F_A(y) dy \leq \int_a^x F_B(y) dy$ for all x with strict inequality holding for at least one x.

[2]

- b) The first order stochastic dominance establishes dominance based on return while second order stochastic dominance identifies the "less volatile" asset. [1]

iii)

Probability of Return/ Return	Return				
	-3.00%	-2.00%	0.00%	2.00%	4.00%
Asset A	0.1	0.2	0.4	0.2	0.1
Asset B	0.2	0.3	0.1	0.3	0.1
Asset C	0.1	0.3	0.2	0.3	0.1

CDF

Asset A	0.1	0.3	0.7	0.9	1
Asset B	0.2	0.5	0.6	0.9	1
Asset C	0.1	0.4	0.6	0.9	1

Integral of CDF

Asset A	0.1	0.4	1.1	2	3
Asset B	0.2	0.7	1.3	2.2	3.2
Asset C	0.1	0.5	1.1	2.0	3

	First order	Second Order
Assets A and B	none	A over B
Assets A and C	none	A over C
Assets B and C	C over B	C over B

[6]
[13 Marks]

Solution 2:

Expected percentage return on the market = sum of the dividend yield of the market and its price appreciation [1]

1.52 % dividend has to be added to appreciation to get expected market return [0.5]

$$E(R_m) = 0.0152 + (15500 - 14370)/14370 = .09384 \quad [1]$$

Using CAPM,

$$E(R_i) = 0.0514 + 1.14 * (.09384 - 0.0514) = .09978 \quad [1.5]$$

If x is the expected price of the stock next year, then the stock return should follow;

$$X = (1 + .09978) * 80 - 2 \quad (\text{where 2 is the dividend amount}) \quad [1.5]$$

This gives x = Rs 85.98

[0.5]
[Max 6 Marks]

Solution 3:

$$i) \quad \mu = \int_{0.5}^{\infty} t f(t) dt = \int_{0.5}^{\infty} 0.375 t^{-3} dt = 0.375 * \left[\frac{t^{-2}}{-2} \right]_{t=0.5}^{\infty}$$

Substituting limits for t, $\mu = 0.75$

Downside semi variance

$$\begin{aligned} \text{Formula} &= \int_{-\infty}^{\mu} (t - \mu)^2 f(t) dt, \\ &= \int_{0.5}^{0.75} (t - 0.75)^2 0.375 / t^4 dt, \\ &= \int_{0.5}^{0.75} t^{(-2)} dt - 2 * 0.75 * \int_{0.5}^{0.75} t^{(-3)} dt + (0.75)^2 \int_{0.5}^{0.75} t^{-4} dt \\ &= 0.02083 \quad (\text{for final result upto 4 decimal places}) \end{aligned}$$

[5]

- ii) (-) Semi-variance is not easy to handle mathematically and it takes no account of variability above the mean. [1/2]
 (-) If returns on assets are symmetrically distributed semi-variance is proportional to variance. [1]
 (-) As with variance of return, semi-variance does not capture skewness or kurtosis. [1/2]
 (+) It takes into account the risk of lower returns. [1/2]
 (+) It can be decomposed into systematic and non-systematic risk contributions. [1/2]

[3]
[8 Marks]

Solution 4:**i) Types of credit events :**

A credit event is an event that will trigger the default of a bond and includes the following:

- actions that are associated with bankruptcy or insolvency laws
- rating downgrade of the bond by a rating agency such as Standard & Poor's or Moody's
- failure to pay
- repudiation/moratorium
- restructuring – where the terms are changed to become less favourable to the bond holder.

[Max 2]

ii) Under the Merton model, the shareholders in the company receive a payoff after 3 years equivalent to that from a call option with strike price equal to the amount to be repaid to the bondholders. [0.5]

The current value of the shareholding can be assessed using the Black-Scholes formula for the value of a call option, with parameters:

$S = 110$, $K = 85$, $\sigma = .3$, $\text{time} = 3$, $r =$ to be estimated as below: [0.5]

Yearly Spot rate to be applied for 3 years = 5.5% (using forward to spot conversion formula) [0.5]

Continuously compounded rate = $\log(1 + 5.5\%) = 5.353\% = r$ [0.5]

Using the above information

E_0 represent the value of the shareholding at time 0:

$$E_0 = 110 * \Phi(d_1) - 85 * e^{(-.05353 * 3)} * \phi(d_2) \quad [1]$$

$$d_1 = 1.0650 \quad [1]$$

$$d_2 = 0.5454 \quad [1]$$

$$\text{Giving } \Phi(d_1) = 0.85657 \quad [0.5]$$

$$\Phi(d_2) = 0.70726 \quad [0.5]$$

$$E_0 = 43.02 \text{ crores, debt} = 66.98 \text{ crores} \quad [1]$$

[Max 7]

[9 Marks]

Solution 5:**i) Let $Z \sim N(0, 1)$. Then, since $W(t) \sim N(0, t)$ we get** [0.5]

$$E[|W(t)|] = E[|\sqrt{t}Z(t)|] \quad [1]$$

$$= \sqrt{t}E[|Z(t)|] \quad [1]$$

$$= \sqrt{t} \cdot 2 \int_0^\infty \frac{e^{-\frac{z^2}{2}}}{\sqrt{2\pi}} z \, dz \quad (\text{from standard normal integration, since the absolute is used, integration is from 0 to infinity multiplied by 2 times}) \quad [2]$$

$$= \sqrt{t} \cdot 2 \int_0^\infty \frac{e^{-w}}{\sqrt{2\pi}} \, dw \quad (\text{by } w = z^2/2, \, dw = z \, dz) \quad [2]$$

$$= \frac{\sqrt{t} \cdot 2}{\sqrt{2\pi}} * [e^{-w}]_\infty^0 \quad [1]$$

$$= \frac{\sqrt{2t}}{\sqrt{\pi}} \quad [0.5]$$

[Max 8]

- ii) ABC Ltd's price is higher than XYZ Ltd's price at the end of the day would be equivalent to:

$$X(t) > 0 \quad [1]$$

If $X(t)$ is an Arithmetic Brownian motion process

Then

$$E[X_T] = x + \mu T \quad [0.5]$$

Where,

$$x = X(3/4) = 40.25 - 39.75 = 0.50 \quad [1]$$

$$\mu = 0 \quad [0.5]$$

$$T = (1 - \frac{3}{4}) \quad [0.5]$$

$$\text{Var}[X_T] = \sigma^2 T = 0.3695/4 = 0.092375 \quad [0.5]$$

$$P(X(1) \geq 0) = P\{(X(1) - E[X_T]) / \sqrt{\text{Var}[X_T]} \geq (0 - 0.5) / \sqrt{0.092375}\} \quad [3]$$

$$= 1 - P(Z \leq -1.6451) = 0.95 \quad [1]$$

[Max 8]

[16 Marks]

Solution 6:

i)

- Black Scholes formula [½]

$$f(t, St) = St \varphi(d1) - ke^{-r(T-t)} \varphi(d2)$$

- Substituting values S_0, K, r, T [1]

$$f(0, S_0) = 11000 * \varphi(d1) - 12000e^{-0.02(1-0)} \varphi(d2)$$

- Calculating $d1, d2$ - correct formula and substitution [2]

$$d1 = (\log\left(\frac{S_0}{K}\right) + (r + 0.5\sigma^2)(T - t)) / \sigma\sqrt{T - t}$$

$$d1 = (\log\left(\frac{11000}{12000}\right) + (0.02 + 0.5\sigma^2)) / \sigma$$

$$d2 = d1 - \sigma$$

- Guessing and Interpolation - demonstration of steps of at least 1 calculation with chosen sigma [3]

sigma	d1	d2	Phi(d1)	phi(d2)	f(0, S ₀)
5%	-1.31523	-1.36523	0.0942	.086091	23.7512
10%	-0.62011	-0.72011	0.2676	0.2357	170.788
15%	-0.37174	-0.52174	0.355	0.300	365.8719

With interpolation the final sigma = 7.96%.

- Final answer $\sigma = 7.96\%$ [1]
Give marks for sigma in the range of 7.2 to 8%

[Max 5]

ii)

- Options are priced by relative valuation techniques (i.e. risk neutral valuation). [1/2]
- This approach is equivalent to building a hedging strategy for the option and does not take account of the expected return on the share. [1]

- Since the hedging strategy involves holding some shares, the drop in price will result in a drop of the value of the option even though the expected future share price has remained the same.

[1]

[Max 2]

iii)

- Call options with lower strike are more valuable $A > C$
- American calls more valuable than European, if they have same strike and expiry, $A > D$,
- $B > E$
- American calls with longer expiry are more valuable, $A > B$
- Possible to clearly rank $A > B > E$, others not so straight forward.

[5]

iv)

Factors affecting option price	American call	American put
Share price	Increase	Decrease
Exercise price	Decrease	Increase
Time to expiry	Increase	Increase
Volatility of share price	Increase	Increase
Risk free rate of interest	Increase	Decrease

[5]

[17 Marks]**Solution 7:**

Ruin theory: calculating premium loading

$$\text{Mean claim} = (50,000 * 0.3 + 1,00,00 * 0.5 + 2,00,00 * 0.2) = 1,05,000$$

[0.5]

$$E(X^2) = 13750000000$$

[0.5]

$$\text{Mean aggregate claim} = 25 * 105000 = 2625000$$

[1]

$$\text{SD of aggregate claims} = \text{SQRT}(25 * 13750000000) = 586301.97$$

[1]

Normal Approximation

$$P(2625000, 586302^2) > 240000 + 2625000 * (1 + \Theta) = 0.1$$

[1]

$$P((N(0,1) > ((240000 + 2625000\Theta) / 586.3)) = 0.1$$

[1]

$$(240 + 2625\Theta) / 586.302 = 1.2816$$

[1]

$$\Theta = 0.1948$$

[Max 6 Marks]**Solution 8:**

i)

a) Risk neutral probability measure –

Q is the *risk-neutral probability measure*, which gives the risk-neutral probability q to an upward move in prices and $1-q$ to a downward move.

[1]

The risk-neutral probabilities ensure that the underlying security yields an expected return equal to the risk-free rate.

[1]

Under the probability measure Q , investors are neutral with regard to risk, they require no additional return for taking on more risk. [1]

[3]

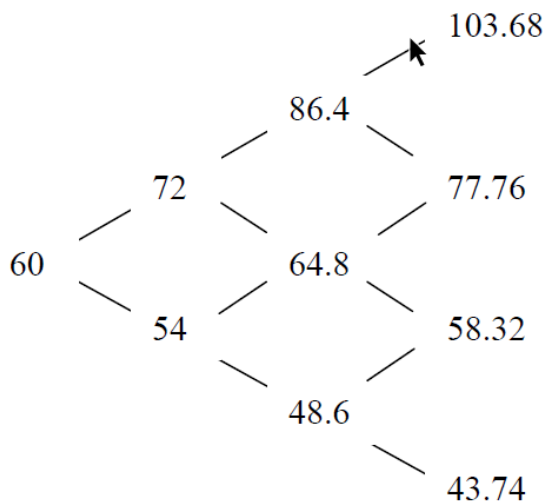
- b) Recombining binary tree A recombining binomial tree (or binomial lattice) is one in which values of u and d , and consequently the risk-neutral probabilities, are the same in all states. [1]

With such models:

- the volume of computation required is greatly reduced
- Nt , the number of up-steps up to time t , has a binomial distribution with parameters t and q [1]

[2]

ii)



$$C_{uuu} = (u^3 S_0 - K)^+;$$

$$C_{uud} = (u^2 d S_0 - K)^+;$$

$$C_{udd} = (u d^2 S_0 - K)^+;$$

$$C_{ddd} = (d^3 S_0 - K)^+.$$

At time 2, we get in the upper state,

$$C_2(uu) = \frac{1}{1+r} [q C_{uuu} + (1-q) C_{uud}],$$

in the medium state,

$$C_2(ud) = \frac{1}{1+r} [q C_{uud} + (1-q) C_{udd}]$$

and in the lowest state,

$$C_2(dd) = \frac{1}{1+r} [qC_{udd} + (1-q)C_{ddd}],$$

where the risk-neutral probability of an upward move is

$$q = \frac{(1+r)-d}{u-d}.$$

At time 1, we get in the upper state,

$$C_1(u) = \frac{1}{1+r} [qC_2(uu) + (1-q)C_2(ud)],$$

and in the lower state,

$$C_1(d) = \frac{1}{1+r} [qC_2(ud) + (1-q)C_2(dd)].$$

At time 0,

$$C_0 = \frac{1}{1+r} [qC_1(u) + (1-q)C_1(d)].$$

$$Q = (\exp(r) - d) / (u-d),$$

Replacing $\exp(r)$ by its Taylor series expansion and ignoring higher order terms, $\exp(r)$ can be replaced by $(1+r)$.

Intermediate answers are provided for both

Formula		Using $(1+r-d)/(u-d)$	Using $(e^r-d)/(u-d)$
Cuuu	$u^4S_0 - K$	43.68	43.68
Cuud	$u^3dS_0 - K$	17.76	17.76
Cudd	$ud^2S_0 - K$	0	0
Cddd	$d^3S_0 - K$	0	0
q		0.7	0.7209
1-q		0.3	0.2791
C2(uu)	$1/(1+r)((qC_{uuu} + (1-q)C_{uud}))$	32.35	32.65
C2(ud)	$1/(1+r)((qC_{uud} + (1-q)C_{udd}))$	11.2	11.47
C2(dd)	$1/(1+r)((qC_{udd} + (1-q)C_{ddd}))$	0	0
C1(u)	$1/(1+r)((qC_{2uu} + (1-q)C_{2ud}))$	23.43	23.69
C1(d)	$1/(1+r)((qC_{2ud} + (1-q)C_{2dd}))$	7.06	7.23
C0	$1/(1+r)((qC_{1u} + (1-q)C_{1d}))$	16.68	16.89

[8]

[13 Marks]

Solution 9:**i) Estimating IBNR**

Dev factors –for consistent correct factors (either weighted or simple averages) Example calculation below has weighted average.

Dev factors	1.279973	1.132151	1.04398
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Cumulative triangle – correct triangle

UWY/ DY	0	1	2	3
2018	10908	13802	15757	16450
2019	9918	12751	14305	14934.14
2020	10102	13034	14756.45	15405.45
2021	9679	12388.86	14026.05	14642.93

Incremental triangle

UWY	0	1	2	3
2018	10908	2894	1955	693
2019	9918	2833	1554	629.1404
2020	10102	2932	1722.453	648.9955
2021	9679	2709.857	1637.197	616.8722

Expected future claims = sum of highlighted cells in above incremental triangle = 7964.5159
[5]

ii) AvE

Use cumulative triangle

Retain first column apply factors to develop the claims as below

UWY	0	1	2	3
2018	10908	13961.94	15807.03	16502.23
2019	9918	12694.77	14372.39	
2020	10102	12930.29		
2021	9679			

Incremental triangle from the expected numbers above

UWY	0	1	2	3
2018	10908	3053.944	1845.082	695.2002
2019	9918	2776.771	1677.624	
2020	10102	2828.286		
2021	9679			

AvE = difference of original incremental vs expected incremental triangle

UWY	0	1	2	3
2018	0	-159.9437	109.9181	-2.200152
2019	0	56.22937	-123.624	
2020	0	103.7144		
2021	0			

[4]

- iii) Any relevant comment on the numbers in the table above such as
- The differences are nearly 10% of the amounts and look reasonable overall.
 - The differences are neither fully positive or negative that implies we are neither under- nor over-estimating consistently. [1]
- iv) BCL assumptions
- The first accident year is fully run off
 - Claims in each development years are a constant proportion in monetary terms of the total claims for each accident year
 - Inflation is not allowed for explicitly, rather it is allowed for implicitly as a weighted average of past inflation. [2]

[12 Marks]
